

LINK.AI

Product Specification

Agent Lead Engine — strategic pivot, phased roadmap, and technical spec

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HOW TO USE THIS DOCUMENT

This document defines the next phase of Link.ai. It is the source of truth for what we are building, why, in what order, and against which APIs. It supersedes prior pitch and roadmap materials where they conflict.

Wireframes will be produced as a follow-up deliverable once the engineering team has reviewed and signed off on the feature spec and phasing herein.

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1. Executive Summary

Link.ai is being repositioned from a consumer-facing real estate search assistant into an agent-led lead engine that captures buyer and seller leads on behalf of any agent — Link agents and outside agents alike — and routes those leads silently into Link's mortgage and escrow back-end where the company captures the bulk of its margin.

Why we are pivoting

Competing with Zillow, Redfin, Homes.com and Realtor.com for consumer search traffic is structurally disadvantageous. Those platforms have eight-figure marketing budgets, decades of SEO equity, and an MLS data parity Link cannot match for a comparable spend. Continuing to fight that battle burns capital on the wrong axis.

Meanwhile, the actual unit economics of a closed Link transaction are heavily skewed toward services: on a representative \$1M California deal, brokerage nets approximately \$2,500, mortgage nets approximately \$4,000, and escrow nets approximately \$6,000. Escrow alone is more profitable than brokerage and mortgage combined. The strategic question is therefore not "how do we win consumer search," but "how do we get more deals — anyone's deals — to close through our mortgage and escrow."

What we are building

A white-labeled, link-and-QR-driven AI tool that any real estate agent can deploy to their existing marketing surfaces — Luxury Presence websites, MLS comments, yard signs, business cards, social bios, email signatures. The agent gets a free or low-cost AI assistant that answers buyer questions, generates home valuations, qualifies buyers, and produces seller net sheets. Every consumer who engages becomes a captured lead in the agent's Link backend, and every consumer who completes a pre-qualification or net sheet is also routed silently to Link's loan officer team and escrow team for follow-up.

The product is genuinely valuable to the agent on its own merits, which keeps the model RESPA-clean (a la Homebot). Link's monetization advantage comes from owning the back-end services rather than from charging the agent or coercing referrals.

Why this works

- Reuses the majority of infrastructure already built — Listing Copilot, FAQs, single-property pages, link/QR manager, home valuation tools, saved searches.
- Attaches the new monetization layer (Arive Zapier handoff, escrow team routing) to flows that capture the most lead intent: pre-qual and net sheet.
- Doubles as a recruitment wedge for Link Brokerages — "join Link, get the AI lead system free; stay independent, pay \$X/month."
- Defensible against pure SaaS competitors. Luxury Presence cannot subsidize tooling with mortgage profits; Link can.

What is in scope for this document

This spec covers the full Phase 1 through Phase 3 build, with detailed flows for the two highest-priority capture features (pre-qualification, which is greenfield, and net sheet, which integrates the existing escrow.link.ai tool into the unified Link.ai surface), API recommendations including soft credit pull and Arive Zapier wiring, the multi-agent reasoning architecture that preserves the

company's NSF research narrative, and an inventory of every requested feature with build status and recommended phase.

THE SINGLE MOST IMPORTANT DECISION

The single most important architectural decision in this spec is to treat the Arive Zapier handoff and the escrow team handoff as silent, post-conversion side effects of the consumer's in-product actions — not as consumer-facing choices, and not as agent-facing requirements. The agent gets the lead in their Link dashboard and (optionally) their CRM. Link's services teams get the lead in parallel. Both happen automatically.

2. Strategic Framing — The Pivot

2.1 The problem with the original direction

The original Link.ai positioning placed the consumer at the top of the funnel: drive consumer eyeballs to Link.ai search, convert them into Link leads, hand them off to a Link agent, close through Link mortgage and escrow. This funnel is intuitive, but it has three structural problems.

1. Top-of-funnel consumer acquisition in real estate is dominated by Zillow, Redfin, Homes.com, and Realtor.com. These incumbents spend hundreds of millions on demand generation annually, have decades of SEO equity, and own MLS data syndication relationships at scale. A capital-efficient startup cannot win this fight on the same axis.
2. Even when consumer leads do come in, they have to be routed through a single Link agent who may or may not provide a great experience, and who is geographically constrained. Lead-to-close conversion is brittle.
3. The brand-recognition battle (consumers learning to type "link.ai" when they want to search homes) is enormously expensive and slow. The pitch deck's SOM of \$43.5M in CA assumes 5% market capture, which is reasonable as a long-term target but not a 12-month wedge.

2.2 The unit-economics realignment

When we examine where the money actually is on a closed deal, the picture changes:

Vertical	Net per deal (\$1M)	Share of total	Strategic implication
Brokerage	\$2,500	~19%	Smallest absolute margin; commoditized; high agent split pressure.
Mortgage	\$4,000	~31%	Healthy margin; the qualifier — gets us to the close.
Escrow	\$6,000	~46%	Largest single bucket. The actual prize.
Total Link net	\$12,500	100%	Closing through any vertical builds the relationship for the next.

Escrow is the single most profitable vertical, and it is also the vertical with the lowest agent loyalty — agents and consumers rarely have strong opinions about which escrow company a deal closes through. This makes escrow simultaneously the most valuable and the most capturable vertical in the stack.

Mortgage sits in the middle. It is the connective tissue: a buyer who pre-qualifies with a Link-network lender is overwhelmingly likely to close that mortgage with Link, and that mortgage close pulls the escrow naturally. So the funnel we actually want to build is:

THE NEW FUNNEL

Agent shares Link.ai widget → Consumer engages with AI → Consumer pre-qualifies (mortgage capture) → Consumer goes under contract through any agent → Escrow opens with Link (escrow capture).

The agent owns the consumer relationship. Link owns the services relationship. Both are happy.

2.3 Why the agent is the right buyer

The agent is the right buyer of this product for four reasons:

- **Agents already control the services decision.** In a typical transaction, the buying agent recommends a lender and the listing agent recommends an escrow company. If we make the agent successful, they bring us the services.
- **Agents already control the marketing surfaces.** Every agent has a website, social presence, business card, MLS listings, and yard signs. Embedding our link/QR into surfaces they already own is dramatically cheaper than building consumer-facing brand recognition from zero.
- **Agents are vocally underserved.** After 3,000 agent interviews and onboarding 250 agents, the team knows that agents repeatedly ask for tools that solve specific workflow pain — pre-qual intake, listing FAQs, single-property pages, social content. None of the existing CRMs (Follow Up Boss, kvCORE, GHL, LionDesk) bundle these with AI in a way that captures consumer intent automatically.
- **Agent acquisition is concentric.** We start with the 250 Link Brokerage agents (free distribution), expand to outside agents through a paid tier, and use the tool itself as a recruiting wedge to grow the brokerage. Each circle reinforces the next.

2.4 Why this is RESPA-clean

RESPA Section 8 prohibits paying for referrals of settlement services. The risk in any "we give agents a free tool, agents send us business" model is that the tool itself can be construed as compensation for referrals. Two design principles keep us clear:

- **Genuine standalone value:** The agent gets a fully functional AI lead engine whether or not they ever send Link a single mortgage or escrow file. There is no requirement, expectation, or condition. This is the Homebot model.
- **Consumer-initiated services routing:** When a consumer chooses to pre-qualify or chooses to request a net sheet, they are interacting with a Link-branded service inside the product. They are choosing Link's mortgage or Link's escrow because that is the path of least resistance, not because the agent forced them.

COMPLIANCE REMINDER

We are not lawyers. Final RESPA counsel sign-off is required before launch. The intent of this spec is to design a product that is plausibly clean by these principles; the legal review converts plausibility into certainty.

3. Product Vision & Positioning

3.1 One-line positioning

Link.ai is the AI lead engine for real estate agents. One link, every workflow — buyer search, pre-qualification, listing inquiries, home valuation, seller net sheets — captured into the agent's pipeline and the agent's CRM, with optional white-label branding and silent routing of services-eligible leads to Link's in-house mortgage and escrow teams.

3.2 What Link.ai is

- **A widget, not a destination.** Consumers reach Link.ai through agent-controlled links and QR codes, never by typing link.ai into a browser. The product lives wherever the agent lives.
- **A capture system, not a search portal.** Every interaction is engineered to convert anonymous consumer intent into a structured lead with contact info, preferences, and qualification data.
- **A multi-workflow tool, not a chatbot.** Pre-qual, net sheet, valuation, listing FAQ, saved search — each is a discrete workflow that produces a different output. The chat is the connective tissue.
- **A services flywheel, not a SaaS subscription.** Revenue comes primarily from mortgage and escrow closes downstream. Agent subscriptions are a secondary, retention-focused revenue stream — not the core unit economics.

3.3 What Link.ai is not

- Not a Zillow competitor. We do not compete for consumer search traffic.
- Not a CRM. We integrate with the agent's existing CRM via Zapier rather than replacing it.
- Not a website builder. We assume the agent already has a Luxury Presence, Real Geeks, or similar website and we plug into it.
- Not a generic agent productivity suite. The marketing-content features (blog posts, social content, ad management) exist to round out the offering, but they are not the core value proposition and they are de-prioritized accordingly.

3.4 Competitive positioning

The competitive set has shifted. We are no longer competing primarily with Zillow and Redfin. The new comparison set:

Category	Examples	Where Link wins
Agent website providers	Luxury Presence, Real Geeks, Placester	They sell pretty pages. We capture leads inside the page and qualify them.

Category	Examples	Where Link wins
Agent CRMs	Follow Up Boss, kvCORE, LionDesk, GHL	They store and route leads. We generate them and pre-qualify them, then sync to their CRM.
Home valuation tools	Homebot, HouseCanary	They send monthly emails. We do that plus pre-qual, plus listing FAQs, plus single-property pages, in one bundle.
Loan officer tools	Rocket Pro Insight, BeSmartee, Maxwell	They are sold to LOs. We are sold to agents and capture leads upstream of the LO.
AI brokerages / iBuyers	Opendoor, HomeLight, Real	They want to own the consumer relationship. We empower the agent to own it.

3.5 The structural moat

No pure-SaaS competitor can match Link's pricing without operating at a loss, because Link is monetizing services revenue that they cannot access. Luxury Presence cannot subsidize their tool with mortgage profits. Homebot cannot bundle escrow. Follow Up Boss cannot offer free CRM seats financed by lender margin. Link can give the tool away and still profit, which is structurally unavailable to any of them.

The moat is not the AI. The AI is replicable. The moat is the vertical integration: brokerage + mortgage + escrow + AI tool, all under one roof, with the back-end services subsidizing the front-end tool.

4. Three-Layer Architecture

Link.ai resolves into three logical layers. Each has its own scope, its own technical considerations, and its own success metric. Most existing infrastructure maps cleanly to Layer 1 (Distribution) and Layer 2 (Capture). Layer 3 (Routing) is where most of the new work in this spec lives.

4.1 Layer 1 — Distribution (Links, QR codes, embeds)

This is how the product reaches the consumer. The agent never has to send the consumer to "link.ai" as a destination; instead the agent shares one of several link types, each tied to a specific intent.

Link type	Purpose	Where it lives
Generic agent link	Catch-all "talk to my AI assistant" entry point	Email signature, social bio, business card, voicemail
Single-property link	Tied to a specific listing; opens to that property's detail page and AI	MLS comments, yard sign rider, listing flyer, Instagram post
Buyer search link	Pre-loaded with a saved search (e.g., 2BR/Encino/<\$1.5M)	Sent in 1:1 followup to a buyer
Home valuation link	Lead magnet for sellers; produces a valuation report	Postcards, Facebook ads, "what's your home worth" widgets
Pre-qualification link	Direct entry to the pre-qual flow	Lender/agent co-marketing, listing pages
White-label embed	JS snippet for embedding the chat into the agent's own site	Phase 3 — for Luxury Presence pages and similar

Each link is unique, tracked, and tied to the agent. Every consumer who lands via that link is attributed to that agent in the dashboard and in any CRM they sync to. QR codes are auto-generated as PNG and SVG so the agent can drop them onto any print material.

STATUS

Build status: Substantially complete. The Link Manager (with QR generation), the per-agent and per-listing link types, and the unique URL infrastructure are already in production. Net new in Phase 1: pre-qual link type and home-valuation lead-magnet link type.

4.2 Layer 2 — Capture (AI conversation + structured workflows)

When the consumer lands, they enter an AI-first experience that is part chat, part guided form, part structured workflow. The chat is the connective tissue, but most consumer value comes

from the discrete workflows that produce concrete outputs: a list of properties, a valuation report, a pre-qual estimate, a net sheet.

The Layer 2 workflows are:

- Property search and Q&A (live)
- Listing-specific FAQ chat (live, with agent-customizable FAQs per listing)
- Home valuation report (live; monthly recurring email is new in Phase 1)
- Saved buyer search with email alerts (live; needs pre-qual layer attached)
- Pre-qualification intake — new (Phase 1)
- Net sheet generator — live at escrow.link.ai; integration into unified Link.ai UI new in Phase 1
- Affordability calculator with payment breakdown — new (Phase 2)
- Loan program comparison — new (Phase 2)

Across all workflows, the moment a consumer provides contact information (name, email, phone), they become a captured lead. Captured leads flow to Layer 3.

4.3 Layer 3 — Routing (Lead distribution and services capture)

Once a lead is captured, the routing layer decides where the lead goes. Every captured lead does the following automatically:

1. Lands in the agent's Link backend dashboard with full context (which link, which workflow, what they asked, what they qualified for).
2. If the agent has connected a CRM, syncs via Zapier outbound to that CRM as a new lead.
3. If the lead completed pre-qualification, pushes via Zapier into Link's Arive instance, creating a draft loan file owned by the appropriate Link loan officer.
4. If the lead requested a net sheet (or if their property is identified as listed), creates a record in Link's escrow team's pipeline for outreach.

Layers 1 and 2 are agent-facing and consumer-facing respectively. Layer 3 is silent — the consumer is not asked to choose a lender or an escrow company, and the agent is not required to use Link's services. The routing happens because the consumer engaged with a Link-branded workflow and chose the path of least resistance.

ARCHITECTURAL RULE

Architectural rule: Layer 3 routing is post-conversion only. We do not nag the consumer to choose Link mortgage during the chat. We do not require the agent to opt into Link services. The services capture is an automatic consequence of the consumer self-selecting into a workflow whose only natural fulfillment provider in our product is Link. This is what keeps the model RESPA-clean and consumer-friendly.

5. Full Feature Inventory

Every feature requested or implied by the current product, the new direction, and the existing infrastructure is catalogued here with build status, recommended phase, and a one-line description. The Phase column is a recommendation; the engineering team should confirm and adjust based on actual progress on partially-built items.

Status legend: Live = in production today. Partial = substantially built but not feature-complete. New = greenfield build. Phase legend: P1 = next 2-4 weeks. P2 = months 2-4. P3 = months 4-6 and beyond.

Feature	Description	Status	Phase
Property search & Q&A	Conversational search with map, filters, save-to-list. Already core to Link.ai.	Live	P1
Listing copilot + FAQs	Per-listing AI Q&A with agent-customizable FAQs and listing-specific context.	Live	P1
Single-property pages	Per-listing landing page with photos, details, and embedded chat for that listing.	Live	P1
Saved searches & buyer alerts	Buyer signs up for criteria, gets emails on new matches.	Live	P1
Home valuation report	Address-driven valuation with comp logic.	Live	P1
Monthly valuation email	Recurring monthly email to seller leads with updated valuation, market trend, and CTA.	New	P1
Pre-qualification intake	Conversational 1003 capture with soft credit pull, lender matching, conditional pre-qual estimate, Arive Zapier handoff.	New	P1
Net sheet generator	Seller-side estimated proceeds calculator. Already built and live at escrow.link.ai. Phase 1 work is integrating into the unified Link.ai UI and wiring escrow team handoff.	Live	P1
Affordability calculator	Buyer enters income/down payment, gets monthly payment, max purchase price, and program suggestions.	New	P2
Loan program comparison	AI explainer of FHA vs Conv vs VA vs Jumbo etc, with eligibility logic.	New	P2
Link Manager	Generate, name, track, and revoke unique links per agent and per listing. QR PNG/SVG export.	Live	P1
White-label theming	Agent uploads logo, sets accent color, optional custom subdomain. Removes Link branding from consumer view.	New	P1

Feature	Description	Status	Phase
JS embed widget	Drop-in script for embedding the AI chat on the agent's own website (Luxury Presence, etc).	New	P3
Listing manager	Agent dashboard view of their listings with FAQ editor.	Live	P1
Account settings	Agent profile, branding, notifications, integrations.	Live	P1
Agent dashboard / KPIs	Active leads, engaged leads, link views, ad metrics, pipeline summary.	Partial	P1
Notifications	Email opens/clicks, link views, pre-qual submitted, valuation requested, etc. Configurable per channel.	New	P2
CRM sync (Zapier outbound)	Push captured leads to agent's CRM via Zapier triggers.	New	P2
Arive Zapier handoff	On pre-qual submit, push borrower record into Link's Arive LOS.	New	P1
Escrow team handoff	On net sheet completion or detected listing, create record in Link's escrow pipeline.	New	P1
Lender guideline matrix	Structured JSON of Link's ~20 lender programs with eligibility rules.	New	P1
Automated blog posts	AI-generated blog posts on neighborhood, market trend, and listing topics. Agent edit/publish.	New	P2
Social media content	AI-generated post copy and image suggestions for Instagram, Facebook, LinkedIn.	New	P2
Meta ad creation	Wizard for creating Facebook/Instagram ads driving to single-property pages or valuation.	New	P2
Google ads creation	Wizard for Google Search/PMAX ads with similar destinations.	New	P2
Agent onboarding flow	Two-track signup: Link Brokerage agent (free) and outside agent (paid/freemium).	New	P2
Brokerage recruitment surface	Marketing site explaining the AI lead engine as a benefit of joining Link Brokerages.	New	P3

NOTE FOR ENGINEERING

The marketing-content cluster (blog posts, social content, Meta ads, Google ads) is

marked Phase 2 and treated as new work in Link.ai. Earlier exploratory work on these features happened on a separate build outside Link.ai and is not directly portable; the team can mine that work for design and copy patterns but should plan to build clean within Link.ai.

6. Phased Roadmap

The phasing is organized around the escrow flywheel: Phase 1 ships the wedge features that capture services-eligible leads, Phase 2 adds the breadth that justifies the agent paying for the tool, and Phase 3 expands distribution surface area and deepens integrations.

6.1 Phase 1 — Wedge (next 2 to 4 weeks)

PHASE 1 GOAL

Goal: ship the smallest, most coherent product that captures services-eligible leads and gets attributed to the agent. Validate with the existing 250 Link Brokerage agents before going outbound.

Phase 1 deliverables

- **Pre-qualification intake flow** — conversational 1003 capture, iSoftpull soft credit pull, lender matrix matching, conditional pre-qual estimate, Arive Zapier push to Link's LO team. Detailed spec in Section 7.
- **Net sheet generator integration** — the seller-side proceeds calculator already exists and is live at escrow.link.ai. Phase 1 work is to integrate it into the unified Link.ai surface (accessible via agent links), wire the agent dashboard capture, and add the silent escrow team handoff plus the listing-detected variant. Detailed spec in Section 8.
- **Monthly valuation email** — cron-driven recurring email for everyone who has run a home valuation. Includes updated estimate, market trend chart, and CTA to talk to the agent.
- **Agent dashboard with KPIs** — active leads enrolled, engaged leads, link views, pre-quals submitted, valuations requested. Detailed spec in Section 13.
- **White-label theming** — agent uploads logo and sets accent color in account settings; consumer-facing surfaces (single-property pages, chat header, valuation report) honor those settings and remove Link branding from chrome (Link branding remains in legal footers and any service-routing surfaces).
- **Lender guideline matrix** — structured JSON of Link's ~20 lender programs with FICO floors, DTI caps, LTV limits, residency requirements, and program-specific rules. Sourced from existing PDF product guides; see Appendix for schema.

Phase 1 KPI target

50 pre-qualifications submitted via Link.ai in the 30 days following Phase 1 launch, of which at least 5 result in Link mortgage applications and at least 10 result in escrow opens (across all sources, not just Phase 1 leads). The success metric is services pull-through, not Link.ai engagement metrics.

6.2 Phase 2 — Depth (months 2 to 4)

PHASE 2 GOAL

Goal: expand the tool from "wedge that captures services" into "tool the agent actually

uses every day." Add the marketing and content features that round out the agent's daily workflow, plus the breadth of buyer-side calculation tools that turn pre-qual into a portfolio of related conversions.

Phase 2 deliverables

- **Affordability calculator** — buyer enters income, down payment, monthly debts; output is max purchase price, monthly payment breakdown, and a soft handoff to pre-qual.
- **Loan program comparison** — AI explainer of FHA, conventional, VA, jumbo, DSCR, ITIN, etc with eligibility logic against the lender matrix.
- **CRM sync via Zapier outbound** — captured leads push to agent's CRM (Follow Up Boss, kvCORE, Salesforce, HubSpot, GHL) via Zapier triggers on lead created, lead engaged, pre-qual submitted, valuation requested.
- **Notifications system** — email and SMS notifications to the agent on link views, email opens/clicks, pre-qual submitted, valuation requested, new home valuation requested. Configurable per agent per channel.
- **Marketing content cluster** — blog posts, social content, Meta ads, Google ads. New builds within Link.ai; design-pattern reference may come from prior exploratory work done outside the Link.ai codebase.
- **Two-track agent onboarding** — "I am a Link Brokerage agent" (free) and "I am an outside agent" (paid or freemium tier).

6.3 Phase 3 — Surface area (months 4 to 6+)

PHASE 3 GOAL

Goal: take the product off Link.ai-hosted pages and put it everywhere the agent already operates. This is where the widget thesis becomes literal — the chat lives inside the agent's Luxury Presence site, the agent's CRM, and the agent's email signature.

Phase 3 deliverables

- **JS embed widget** — drop-in <script> tag for embedding the chat into any agent website. Auto-attribution to the agent ID encoded in the snippet.
- **Native integrations with Luxury Presence and Real Geeks** — deeper than embed; menu in those platforms' admin to configure Link.ai directly.
- **Brokerage recruitment surface** — marketing site that shows the AI lead engine as a benefit of joining Link Brokerages, with a self-serve signup path for licensed agents in approved states.
- **Deeper Arive integration** if and when Arive launches a true REST API. Until then, Zapier remains the integration path.
- **Multi-agent (A2A) reasoning rollout** — upgrade the underlying chat from a single-agent OpenAI orchestration to the A2A protocol described in Section 11. Preserves the NSF research narrative and improves accuracy on cross-domain queries (affordability + program eligibility + tax incentives etc.).

7. Pre-Qualification Flow — Detailed Spec

Pre-qualification is the most important new feature in this spec. It is the wedge that captures services-eligible leads, the data point that justifies our Arive routing, and the workflow most likely to convert chat engagement into a closed mortgage. This section specifies the flow end-to-end.

7.1 Goal

Take a consumer from "I want to know what I can afford" to "I have a conditional pre-qualification estimate and a Link loan officer is reaching out" in under five minutes, without the consumer ever leaving the chat. Maintain consumer-friendly framing throughout — this is not a binding application, no SSN required, no commitment.

7.2 Trigger

The pre-qual flow is entered through any of the following:

- Consumer follows a pre-qualification-typed link (URL parameter ?intent=prequal).
- Consumer asks the AI an affordability question ("Can I afford this?", "How much can I qualify for?", "What's the monthly payment on...").
- Consumer clicks a CTA inside the home valuation report ("Get pre-qualified to make an offer").
- Consumer clicks a CTA inside a single-property page ("See if you qualify for this home").
- Consumer asks about a specific loan program (FHA, VA, conventional) — the AI explains and offers pre-qual.

7.3 Step-by-step flow

Step 1 — Soft introduction

AI message: "Happy to give you a sense of what you can qualify for. This takes about 5 minutes, doesn't affect your credit score, and we don't need your social security number. Want to start?" Consumer confirms with a tap.

Step 2 — Loan purpose

Three buttons: "Buying a home", "Refinancing", "Just exploring". The "Just exploring" path drops the consumer into the affordability calculator (Phase 2) instead. The two real paths continue.

Step 3 — Property context (if available)

If the consumer entered the flow from a single-property page, the property is already attached. Otherwise, ask: "Do you have a specific property in mind, or are you shopping?" If they have one, capture address. If shopping, capture target city/state and price range.

Step 4 — Personal info

Ask in conversational chunks (not as a single 1003 form). Order optimized for conversion: legal name → email → phone → date of birth (year only at this stage to gauge age bracket) → current address. Show progress indicator ("2 of 5 sections done").

Step 5 — Income & employment

Employment status (W-2, self-employed, retired, other), employer name, monthly gross income, years at current employer. For self-employed, ask for two-year average. Capture co-borrower income if applicable.

Step 6 — Assets & down payment

Available cash for down payment, breakdown by source (savings, gift, retirement, sale of current home), interest in down payment assistance programs.

Step 7 — Debts & obligations

Monthly debt payments (credit cards, auto, student loans, child support). Skipping is allowed but flagged for the LO.

Step 8 — Credit consent and soft pull

Consent screen with FCRA disclosure. Explicit consumer click. Backend calls iSoftpull with name and address (no SSN required). Receive credit score and tradeline summary in 2-5 seconds.

COMPLIANCE

Compliance: The credit consent language must be reviewed by counsel. iSoftpull provides recommended FCRA-compliant disclosure copy; we should adopt that and have legal sign off before launch.

Step 9 — Lender matching

Backend runs the captured 1003 + soft credit data against the lender guideline matrix. Output is a ranked list of programs the consumer is eligible for, with estimated rate, monthly payment, and any conditions (PMI required, gift-funds restriction, etc.).

Step 10 — Result presentation

AI returns a conversational summary in chat, e.g.:

SAMPLE AI RESPONSE

"Based on what you've told me, you likely qualify for a 30-year conventional loan up to about \$612,000, with a monthly payment of around \$4,180 at today's rates. You also qualify for FHA, which would let you put as little as 3.5% down — though it adds mortgage insurance.

A few notes: this is a conditional estimate based on a soft credit check, not a binding pre-approval letter. To get an official letter you can include with an offer, one of our loan officers will reach out within 1 business day. We also work with Coby Chatwin (your agent) and your loan officer can coordinate directly with him."

Step 11 — Backend routing (silent)

On submission of the pre-qual, the backend simultaneously:

- Creates the lead in the agent's Link dashboard with full context.
- If the agent has CRM sync enabled, fires Zapier outbound trigger (new lead, qualified).

- Fires Zapier outbound to Arive: creates draft loan file with completed 1003 fields, assigns to the appropriate Link LO based on state and load balancing.
- Sends notification email to the agent: "New pre-qualified lead from your Link.ai assistant."
- Triggers email to the consumer with a copy of the pre-qual summary and contact info for both their agent and their assigned Link LO.
- Logs full event chain (link source, intent, workflow, soft pull result, lender match, routing) for analytics and compliance.

7.4 Edge cases

- **Soft pull fails** (API error, identity mismatch, frozen file). Show consumer "We couldn't pull credit automatically — that's normal, our LO can run it manually. Submitting anyway." Push the lead to Arive without credit data; the LO handles the manual pull.
- **No matching lender program** (very low FICO, very high DTI, etc). Show consumer "Based on what we have so far, none of our standard programs are an exact fit, but our LO can review your situation in detail — there may be programs we can match you with that need a closer look." Push to Arive flagged for high-touch review. Never tell a consumer they "don't qualify"; always defer to the LO.
- **Consumer abandons mid-flow**. Save partial state. After 24 hours, send "Want to finish your pre-qualification?" email. After 7 days, push partial lead to the agent dashboard as "incomplete pre-qual" so the agent can follow up directly.
- **Cross-state agent** (consumer in TX, agent in CA). Route to a Link LO licensed in the consumer's state, not the agent's. Agent dashboard shows lead but flags state mismatch.
- **Consumer is already in Arive** (repeat lead). Detect via email match in the Zapier de-duplication logic. Update existing record rather than creating duplicate.

7.5 Data captured

A complete pre-qual session produces the following structured record. This is the schema that flows into the agent dashboard, the CRM, and Arive.

Field	Source	Routing destination
Full name, email, phone, DOB year, address	Step 4	Dashboard, CRM, Arive
Property address or target search criteria	Step 3	Dashboard, CRM, Arive
Loan purpose, occupancy	Step 2/3	Arive (loan file primary)
Monthly income, employer, employment status	Step 5	Arive 1003, dashboard summary
Assets, down payment	Step 6	Arive 1003

Field	Source	Routing destination
source		
Monthly debt obligations	Step 7	Arive 1003
Credit consent timestamp + IP	Step 8	Compliance log only
FICO score, tradeline summary	iSoftpull soft pull	Arive (locked, LO-only access)
Matched programs, estimated rate, payment	Lender matrix engine	Dashboard, consumer email, Arive notes
Source link / agent ID / referrer	Link Manager	Dashboard for attribution

8. Net Sheet Flow — Integration Spec

The net sheet is the seller-side counterpart to pre-qual. It is the moment a homeowner stops daydreaming and starts running numbers. Capturing this intent is the cleanest signal of "this person is going to list within the next 90 days." Routing it to the escrow team turns the highest-margin vertical into a default outcome.

ALREADY BUILT — INTEGRATION ONLY

The net sheet calculator itself is already built and live at escrow.link.ai. This section therefore specifies an integration, not a build. The Phase 1 work is to (a) make the existing calculator reachable via agent-distributed Link.ai links so the lead is attributed to the agent, (b) capture the resulting record in the agent dashboard and Link's services pipeline, and (c) add the listing-detected variant that fires when a previously-valued home is detected as newly listed. Anything in this section that conflicts with how the live tool already works should defer to the live tool; flag conflicts and we'll align before wireframes.

8.1 Goal

Take a homeowner from "I'm thinking about selling" to "Here is exactly how much I would walk away with at \$X sale price" using the existing calculator, with the resulting record routed silently to Link's escrow team for outreach and attributed to the agent whose link drove the visit.

8.2 Trigger

How a consumer enters the net sheet flow within the unified Link.ai surface:

- Consumer follows a net-sheet-typed agent link (which deep-links to the existing calculator with agent attribution attached).
- Consumer asks the AI a seller-related question ("How much would I net if I sold my house?", "What are closing costs for sellers?") and the AI hands off to the calculator.
- Consumer clicks a CTA inside the home valuation report ("See your estimated proceeds at this price").
- Detected listing event (a consumer's previously-valued home appears on MLS as a new listing). Triggers an outreach email to the consumer offering an updated net sheet, plus an internal alert to the escrow team in parallel.

8.3 Integration touchpoints

Rather than re-spec the calculator step-by-step, this section lists the touchpoints where the existing calculator needs to read or write data into the broader Link.ai system. Engineering should map each touchpoint against the current escrow.link.ai implementation and call out anything that requires changes there.

Touchpoint 1 — Agent attribution on entry

When a consumer arrives via an agent link, the calculator must capture and persist the `agent_id` and `source_link_id` that drove the visit. URL parameter convention: ?

agent={agent_id}&link={link_id}. Persist these on every record produced so attribution survives all downstream routing.

Touchpoint 2 — Pre-fill from prior valuation

If the consumer has previously run a home valuation through Link.ai (and we have their address), pre-fill the calculator's address and estimated value fields. Reduces friction and lifts completion rate. Lookup keyed by consumer email or session ID.

Touchpoint 3 — Lead capture (email/phone)

When the consumer saves or emails the result, capture email and phone, write a record to the canonical netSheet collection (see Section 9.2 data model), and emit a lead-created event to the routing layer. The existing tool may already do this; the integration ensures the record lands in Link.ai's shared lead store rather than only the standalone tool's store.

Touchpoint 4 — Branded PDF output

The PDF the calculator produces should honor the white-label theme of the agent who drove the visit (logo, accent color, agent name and contact info in footer). If the existing PDF generator is templated, this is a template variable injection; if it is hardcoded to Link branding, this is a refactor. Confirm with the implementation team.

Touchpoint 5 — Silent escrow handoff

On record completion (consumer saves/emails the net sheet), the routing layer fires the following side effects:

- Creates the lead in the agent's Link dashboard with the net sheet attached as a viewable / downloadable artifact.
- Pushes via Zapier outbound to the agent's CRM if connected (Phase 2).
- Creates a record in Link's escrow team pipeline tagged "net sheet completed; expected listing within 90 days."
- Sends notification email to the agent.
- Sends the consumer the branded PDF.

8.4 Listing-detected variant (new in Phase 1)

When a previously-valued home appears on MLS as a new active listing — regardless of which agent listed it — the system detects the match and triggers a parallel flow:

- Email to the consumer: "We noticed your home was just listed. Here's an updated net sheet at the asking price." (Links back into the existing calculator pre-filled with the listing price.)
- Internal alert to Link's escrow team: "Listed property detected — escrow opportunity."
- No alert to the original Link agent if the listing is held by another agent (avoids awkward poaching dynamics).

WHY THIS MATTERS

The listing-detected variant is the single highest-value automated signal we will produce. A net sheet completion is intent; an actual listing event is action. Routing this to escrow within minutes of MLS posting puts Link's escrow team ahead of every competitor by

definition.

8.5 Data captured

Each completed net sheet produces a record similar in shape to pre-qual but seller-side. Fields captured: agent_id, source_link_id, property address, sale price assumption, mortgage payoff entered, fee breakdown, computed proceeds, consumer name/email/phone, timestamps. Routing destinations: agent dashboard, agent CRM (Phase 2), Link escrow pipeline.

9. Technical Architecture Additions

Link's current stack — Next.js + React + TypeScript on the frontend; LoopBack 4 + MongoDB on the backend; OpenAI, Google Maps, Tavily, DocuSign, CometChat, Replier on the integration side; Auth0, Docker, Terraform, AWS for infra — is the right shape for what we are building. The pivot does not require replatforming. It requires adding a small set of new services and capabilities, and refactoring the AI layer to support multi-agent orchestration over time.

9.1 Net new services

Service	Purpose	Phase
Soft credit pull (iSoftpull)	Real-time credit score and tradeline data with name+address only.	P1
Lender matrix engine	JSON-config of 20 Link lender programs + matching logic.	P1
Arive Zapier outbound	Pushes pre-qual records into Link's Arive instance.	P1
Escrow pipeline router	Internal service that creates records in escrow team's queue on net sheet / listing detected events.	P1
Net sheet calculator	Already exists at escrow.link.ai. Phase 1 work is integration into Link.ai surface, agent attribution, white-label theming, and shared lead store.	P1
Property data enrichment (ATTOM or Estatic)	Tax records, ownership history, mortgage info — fills gaps in MLS data and powers listing detection.	P1/P2
Doc generation (PDF)	Net sheets, pre-qual summaries, valuation reports as branded PDFs.	P1
Notifications service	Email + SMS notifications to agents and consumers, configurable.	P2
CRM Zapier outbound	Pushes captured leads to agent's connected CRM.	P2
MLS listing-event watcher	Polls or webhooks MLS for new listings on previously-valued addresses.	P2
White-label theming layer	Per-agent branding config applied to consumer-facing surfaces.	P1
A2A orchestration layer	Multi-agent reasoning replacing single-agent OpenAI calls.	P3 (research-track)

9.2 Data model additions (MongoDB)

New collections (or extensions to existing collections) needed to support the spec:

- **preQualRecord** — a structured 1003-shape document with embedded credit summary, matched programs, routing log, and Arive sync state. Indexed on agent_id, consumer_email, created_at.
- **netSheet** — property reference, sale price assumption, mortgage payoff, fee breakdown, computed proceeds. Indexed on agent_id, property_address.
- **lenderProgram** — normalized lender + program with eligibility rules (FICO_min, DTI_max, LTV_max, occupancy, residency, special_conditions[]). See appendix for full schema.
- **agentBranding** — per-agent theme settings (logo URL, accent color, custom subdomain, footer disclaimer text).
- **leadRoutingEvent** — immutable log of every lead-to-services routing event. Used for compliance, attribution, and analytics. Records: lead_id, source_link, workflow_completed, routed_to[], timestamp.
- **detectedListing** — records of previously-valued properties detected as newly-listed on MLS. Used to fire the listing-detected escrow handoff.

9.3 AI layer evolution

The current AI layer is a single-agent OpenAI orchestration with retrieval-augmented generation against MLS, valuation, and listing data. This works for Phase 1 and Phase 2. The Phase 3 evolution is to refactor into a multi-agent A2A architecture as described in Section 11. The refactor is non-blocking for Phase 1 and 2 — those phases ship on the current architecture.

9.4 SSE / real-time

The existing Server-Sent Events pipeline for streaming chat responses remains the right primitive. The new flows (pre-qual, net sheet) should reuse it for streaming AI responses and progress indicators (e.g., "running soft credit pull..."). For backend-to-frontend events that are not chat-driven (Arive sync confirmations, listing detected alerts, agent dashboard updates), the existing SSE infrastructure can carry those too with minor extensions to the event taxonomy.

9.5 Compliance & audit

Three areas where the architecture has compliance load:

- **FCRA (credit pulls)**. Every soft pull must be preceded by an explicit consumer consent click logged with timestamp, IP, and consent text version. Credit data must be stored encrypted at rest and access-restricted to LO roles.
- **GLBA (consumer financial data)**. Pre-qual records contain non-public personal information. Encryption at rest and in transit, with role-based access controls and audit logging on read access.
- **RESPA (services routing)**. The leadRoutingEvent log is the audit trail showing that services routing happened post-conversion based on consumer-initiated workflow completion, not as a condition of using the tool. This log should be preserved indefinitely.

10. API & Integration Recommendations

The following external services are recommended for the build. Each entry includes the rationale, the integration model, and notes on alternatives the engineering team should briefly evaluate before committing.

10.1 Soft credit pull — iSoftpull (primary) / Soft Pull Solutions (alt)

DECISION

Recommendation: iSoftpull, with Soft Pull Solutions as a backup option to evaluate during integration.

Why iSoftpull

- Soft pulls priced at approximately \$3.49 per bureau — economically negligible against a \$4K-\$6K services net per closed deal.
- Requires only name and address — no SSN or DOB required to pull. Critical for chat-based UX where asking for SSN tanks conversion.
- Three API products: Intelligence Indicator (decision-only), Report & Score View (full report), Full Feed JSON (storage in our own infra). We use Full Feed for our purposes.
- Native Zapier support if we ever need it for backup integration paths.
- Single agreement covers Experian, TransUnion, and Equifax.

Integration model

Direct REST API call from our LoopBack backend during pre-qual Step 8. Response in 2-5 seconds. Store the full feed encrypted in the preQualRecord document, with FICO and tradeline summary surfaced to the matching engine. Never expose the full credit report to the consumer; only summary attributes ("FICO 720+, no late payments in 24 months").

Open question

Do we want tri-merge (all three bureaus) or single-bureau on first pull? Tri-merge costs ~\$10.50 per pull vs \$3.50 for single bureau. Recommendation: single-bureau (TransUnion) for first pass, with tri-merge available as an LO-side action when the lead is being formally underwritten. This keeps unit economics tight on speculative consumer leads.

10.2 LOS handoff — Arive (Zapier-based)

Arive is the company's LOS today. Arive does not currently expose a true REST API for inbound integrations; the only supported integration model is Zapier. This is a real architectural constraint we need to design around.

What this means

- We can push borrower records into Arive via Zapier — creating draft loan files with completed 1003 fields.
- We cannot synchronously call Arive from our chat UI and get back live pre-approval logic. The "5-minute pre-qual" experience is delivered by our own lender matrix engine

(Section 7); the Arive push is an asynchronous side effect that hands the warm lead to our LO team for the official approval letter.

- For status updates flowing back from Arive (loan stage, document needs, conditions), Zapier triggers fire on milestone events. These can be ingested into our system via webhook to display in the agent dashboard.

Integration model

Phase 1: outbound only. On pre-qual completion, fire a Zap that creates a borrower in Arive with mapped 1003 fields, assigned to the appropriate Link LO. Phase 2: inbound milestone events from Arive feeding loan-status updates back into the agent dashboard. Phase 3: re-evaluate if Arive has launched a true REST API or if we should consider migration to Encompass / LendingPad / similar.

10.3 Property data enrichment — ATTOM Data or Estated

For features that need authoritative property data beyond MLS (tax history, ownership chain, mortgage liens, public records), we need a property data provider. Two options to evaluate:

- **ATTOM Data** — deep property data including tax assessment, ownership, mortgage history. Stronger for valuation and net sheet calculations. Higher cost.
- **Estated (now part of First American)** — similar coverage, often more cost-effective for high-volume use.

Recommendation: have engineering team request pricing and sample data from both, then choose based on coverage in California and the cost-per-call ratio at projected volumes.

10.4 MLS data — current provider unchanged

The existing MLS integration (Bridge / RESO Web API or whatever Link is using today) remains the source of truth for active listings, photos, and status. The new requirement is the listing-event watcher that polls or subscribes to MLS feed for new listings matching previously-valued properties in our database. Engineering should confirm whether the current MLS feed contract supports webhook subscriptions or if we need to poll.

10.5 Document generation — Documenso, PDFShift, or custom

We need to generate branded PDFs for net sheets, pre-qual summaries, and valuation reports. Three options:

- **PDFShift / DocRaptor / similar HTML-to-PDF API** — generates from styled HTML, easy to brand per-agent, low cost.
- **Documenso (open-source)** — if we want signature workflows alongside PDF generation. Adds value if pre-qual ever moves to a signed disclosure.
- **Custom server-side (Puppeteer + handlebars)** — full control, no per-document fee, but more infra to maintain.

Recommendation: PDFShift or DocRaptor for Phase 1 simplicity; revisit if signature workflows become required.

10.6 CRM outbound — Zapier (for Phase 2)

Rather than building native integrations to Follow Up Boss, kvCORE, GHL, HubSpot, Salesforce one by one, we use Zapier as the universal outbound layer. We define our triggers (lead created, lead engaged, pre-qual submitted, valuation requested, net sheet completed) and let agents wire those to whichever destination they prefer. Native integrations can come later for the top 2-3 CRMs by adoption.

10.7 Email & SMS — keep current

Whatever Link is using today for transactional email (Replier per the stack) is fine. For SMS notifications to agents (Phase 2), recommend Twilio for reliability and per-agent number support. For the consumer-facing emails (pre-qual summaries, valuation reports, monthly valuation updates), the existing email infrastructure is adequate — what we add is templated content, not new infrastructure.

10.8 Summary table

Capability	Recommended provider	Phase	Notes
Soft credit pull	iSoftpull	P1	Single-bureau on first pull; tri-merge LO-side later.
LOS handoff	Arive (Zapier)	P1	Zapier-only constraint; design accordingly.
Property enrichment	ATTOM or Estated	P1/P2	Choose based on CA coverage and pricing.
MLS feed	Existing provider	P1	Add listing-event watcher.
PDF generation	PDFShift	P1	Revisit if signatures needed.
CRM outbound	Zapier	P2	Native integrations for top CRMs in Phase 3.
Email transactional	Replier (current)	P1	No change.
SMS notifications	Twilio	P2	Per-agent sender numbers.
LLM orchestration	OpenAI (current)	P1/P2	Refactor to A2A in P3.

11. Multi-Agent (A2A) Reasoning Architecture

NSF NARRATIVE CONTINUITY

This section preserves and refines the multi-agent technical narrative from the NSF application. The "free public-information utility" framing is dropped, but the underlying technical innovation — distributed AI reasoning over fragmented, fast-changing real estate data — is even more relevant under the new direction. Pre-qualification reasoning across 20 lender guidelines, dozens of program rules, and live property data is exactly the kind of cross-domain problem A2A was designed for.

11.1 Why multi-agent

A user query like "Can I afford this 2-bedroom in Encino if I put 5% down on an FHA loan?" requires reasoning across at least four distinct domains: financial (loan math, DTI, payment calculation), policy (FHA program rules, county loan limits, MIP requirements), market (current Encino comps, neighborhood appreciation, days-on-market), and regulatory (CA-specific transfer tax, local property tax assessment behavior). A single monolithic LLM can attempt this, but it makes errors at the boundaries between domains and cannot produce explainable, auditable intermediate outputs.

The A2A approach decomposes the query into specialized agents, each with its own data sources, retrieval strategy, and reasoning constraints, and an orchestration protocol that governs how they exchange information and synthesize a final response.

11.2 Agent roster

Agent	Responsibility	Data sources
Finance Agent	Mortgage math, payment calculation, DTI, LTV, affordability bounds.	Lender matrix, current rate sheet, soft pull data.
Policy Agent	Loan program eligibility, FHA/VA/conventional rules, regional loan limits, assistance programs.	Lender program rules, FHFA loan limits, DPA program registry.
Market Agent	Comparable sales, days on market, list-vs-sale spread, micromarket trends.	MLS feed, ATTOM/Estated property data.
Property Agent	Specific property attributes, listing-specific FAQs, tax history, HOA, zoning.	MLS detail, agent-uploaded FAQs, ATTOM tax records.
Regulatory Agent	State and county fees, transfer taxes, recording fees, prorations.	County fee schedules, ALTA title rates.
Orchestration Agent	Decomposes user query, routes to specialists, synthesizes responses, handles consensus on conflicts.	Coordinates the others; consumes their outputs.

11.3 A2A protocol

The communication protocol governs how agents exchange information. Each message between agents carries semantic content (the task or the partial result), provenance (which data source, which agent, when), and a confidence signal. The orchestration agent uses these signals to decide whether to accept a result, request clarification, or invoke a different specialist.

Key protocol primitives:

- Task: a specific question or computation request, with constraints and context.
- Result: a structured answer with provenance and confidence.
- Clarification: a back-channel question from a specialist to the orchestrator if the task is underspecified.
- Conflict: a flag raised when two specialists return inconsistent results, triggering consensus logic.
- Trace: a complete log of the inter-agent exchanges produced for a single user query, used for explainability and debugging.

11.4 Phase rollout

Phase 1 and Phase 2 ship on the current single-agent architecture. The A2A refactor is a Phase 3 research-track effort that runs in parallel with the product builds. The pre-qual flow in Phase 1 already exhibits multi-agent shape internally (lender matrix engine + soft credit + property context) — formalizing this into the A2A protocol is the architectural cleanup that follows.

11.5 Why this is research-grade

The technical novelty is not "we have multiple specialized models" — that pattern is increasingly common. The novelty is in the protocol design for real-time, mutually-grounded reasoning across data sources that are heterogeneous (structured + unstructured), fast-changing (rate sheets update intraday, MLS updates by the minute), and regulatorily sensitive (lender guidelines have legal force). The contribution is the schema for trust signals, conflict resolution, and provenance preservation in distributed reasoning over live data — a substrate that generalizes beyond real estate to any consumer domain with similar characteristics (healthcare benefits, tax planning, immigration, etc.).

This is the angle that keeps the NSF Phase I narrative alive. It is also the architectural foundation that makes the product genuinely defensible against pure-LLM competitors who are structurally limited to single-agent reasoning over static or shallow grounding.

12. White-Label, Branding & Recruitment

12.1 White-label theming

Any outside agent who pays for Link.ai gets a white-labeled consumer experience. Link Brokerage agents get the option of full Link branding (default) or partial white-label (their photo and brokerage logo prominent, Link branding present in footer).

What gets themed

- Header logo on consumer-facing pages and chat surface.
- Accent color (defaults to Link turquoise, agent can override).
- Agent name and photo prominent in chat header and email footers.
- Custom subdomain (Phase 2): agent.theiragency.com pointing at Link.ai with branded chrome.
- PDF outputs (net sheets, pre-qual summaries, valuation reports) carry the agent's branding.

What does not get themed

- Legal disclaimers and footer text remain Link-branded for liability clarity.
- The pre-qual flow's lender brand reveal — when the consumer sees the matched program, the specific lender is named (this is required for consumer transparency).
- The escrow handoff confirmation email — when escrow opens with Link, the consumer is told this clearly.

12.2 Tradeoff: brand-building vs agent acquisition

White-labeling means every consumer who interacts with a great Link.ai experience attributes the experience to the agent, not to Link. This is intentional — it makes agents loyal — but it means we forgo brand-building with consumers. We accept this tradeoff because the unit economics of services capture work without requiring consumer brand recognition. If we later decide to go direct-to-consumer, we will need to reverse this; the architecture should preserve that optionality (default Link branding, agent override is an explicit setting).

12.3 Recruitment positioning

Link.ai is the strongest single recruiting wedge Link Brokerages has ever had. The pitch to a recruited agent is concrete and differentiated:

THE RECRUITING PITCH

"Join Link Brokerages and you get our AI lead engine free, our mortgage and escrow integration that lets you offer up to \$20K in closing credits to your buyers and sellers, and your own backend dashboard to manage everything. Stay independent and you can still use Link.ai as an outside agent — but at \$X/month and without the bundling discounts."

This pitch works because it is true. Link agents do get the tool free. The bundling discounts are real. The brokerage support is real. The recruitment surface (Phase 3) makes this pitch scalable.

13. KPIs, Notifications & Agent Dashboard

13.1 Agent dashboard KPIs

The agent dashboard surfaces a focused set of metrics that align to what makes the agent successful and what tells us the tool is delivering value. Five primary KPIs at the top of the dashboard:

KPI	Definition	Why it matters
Active leads enrolled	Count of leads in the agent's pipeline who have provided contact info via any Link.ai workflow.	The headline lead count.
Engaged leads	Active leads who have interacted in the last 7 days (chat message, email open, link reopen).	Activity signal — separates dead leads from live ones.
Active links	Count of unique links the agent has currently published.	Distribution surface; a low number means the agent isn't marketing.
Link views	Aggregate count of consumer landings on agent links in the last 30 days.	Top of funnel; the input to lead generation.
Ad performance	Spend, impressions, clicks, leads attributed to ads launched through Link.ai (Phase 2).	ROI on paid distribution; only meaningful once Meta/Google ad features ship.

Secondary panels surface workflow-specific metrics:

- Pre-quals submitted (count, conversion rate from chat-to-submission).
- Net sheets generated (count).
- Home valuations requested (count, count of monthly recurring email subscribers).
- Saved searches active (count, count with email opens in last 30d).
- Listing engagement (per-listing chat sessions and FAQ-answer hits).

13.2 Notifications

The agent gets notified when their leads do things. Default channel is email; SMS opt-in available in Phase 2.

Event	Notification	Channel default
Email opened by lead	Aggregate daily digest	Email digest
Email link clicked by lead	Real-time	Email
Unique link viewed	Real-time (single-property pages); daily digest (generic links)	Email

Event	Notification	Channel default
Pre-qualification submitted	Real-time, high priority	Email + SMS (P2)
Net sheet completed	Real-time, high priority	Email + SMS (P2)
New home valuation requested	Real-time	Email
New saved search lead	Real-time	Email
Lead replied in chat	Real-time	Email + SMS (P2)

Notification fatigue is real. Default settings should be conservative — high-priority events real-time, low-priority events digested daily. Each agent can override per channel and per event type.

14. Open Questions for the Dev Team

These are decisions and judgment calls that this spec deliberately does not resolve, because they require engineering input the strategy work cannot provide. Each is flagged for explicit discussion in the spec review meeting.

Architecture & infrastructure

1. Marketing-content cluster (blog posts, social content, Meta ads, Google ads): prior exploratory work happened on a separate build outside Link.ai. Engineering should review what is salvageable as design or copy patterns and confirm Phase 2 build estimate net of any reusable artifacts.
2. Listing-event watcher: does our current MLS feed contract support webhook subscriptions for new listings, or do we need to poll? If polling, what cadence is acceptable to MLS terms of service?
3. White-label theming: how deep do we go in Phase 1? Logo + accent color is straightforward. Custom subdomain requires DNS handling and SSL provisioning per agent — should that be Phase 2?
4. Property data enrichment: ATTOM or Estated? Engineering should request pricing and sample data from both before committing.

Pre-qualification flow

1. Soft credit pull: single-bureau (TransUnion) or tri-merge on first pull? Recommendation in this spec is single-bureau but the LO team should weigh in on whether that gives them enough to work with.
2. Lender matrix construction: do we encode the 20 lender programs as JSON manually from the PDF guides, or do we ingest the PDFs and use AI extraction? If manual, who is the SME owner per program for ongoing maintenance?
3. Cross-state routing: do we have Link LOs licensed in every state where Link Brokerages currently operates? If not, how do we handle pre-qual leads from agents in states without LO coverage?
4. Lead deduplication: if a consumer pre-qualifies through Agent A on Monday and through Agent B on Tuesday, who "owns" the lead in the agent dashboards? Recommend last-touch with notification to first-touch agent, but this needs a policy decision.

Net sheet flow

1. Mortgage payoff data: do we have a consumer-facing way to look up an actual payoff balance (via Plaid or Finicity loan-account data), or do we rely on consumer self-report?
2. State-aware fee schedules: Phase 1 supports California (where 100% of current Link Brokerage agents are). When do we expand fee schedules to additional states?
3. Listing-detected variant timing: how fast can the MLS watcher fire? Same-day detection is the value proposition. If our feed is daily-batched, that needs to change.

Compliance & policy

1. FCRA disclosure language: legal sign-off required before launch on the soft pull consent screen.

2. RESPA review: external counsel review of the Layer 3 routing architecture before launch. This document's claim of "RESPA-clean" is a designer's plausibility claim, not a legal opinion.
3. Data retention: how long do we retain pre-qual records and credit data? GLBA-compliant retention policy needed.
4. Consumer-facing escrow disclosure: when escrow auto-routes to Link, what is the disclosure language? At what point in the flow do we tell the consumer "by the way, we recommend Link Escrow for this close"?

Distribution & growth

1. Two-track agent onboarding: what is the price point for outside agents in Phase 2? Recommend testing free trial → \$99/month after Y leads to align tool cost with services capture economics.
2. JS embed widget (Phase 3): how do we handle authentication and attribution from a third-party site? OAuth handshake or signed JWT in the snippet?
3. Brokerage recruitment surface (Phase 3): does this live on link.ai or on linkbrokerages.com? Affects SEO strategy.

15. Appendix

15.1 Lender matrix schema (proposed)

Each lender program is encoded as a structured document. The matching engine evaluates a consumer's profile against every program and ranks the eligible matches.

LENDERPROGRAM DOCUMENT — SAMPLE

```
{
  "lender_id": "lender_acme_001",
  "lender_name": "Acme Lending",
  "program_id": "acme_conv_30yr_fixed",
  "program_name": "Acme Conventional 30-year Fixed",
  "loan_type": "conventional",
  "term_years": 30,
  "rate_type": "fixed",
  "occupancy": ["primary", "second_home", "investment"],
  "states_eligible": ["CA", "NV", "AZ"],
  "min_fico": 620,
  "max_dti": 50,
  "max_ltv": 97,
  "min_down_pct": 3,
  "loan_amount_min": 100000,
  "loan_amount_max": 766550,
  "pmi_required_below_ltv": 80,
  "first_time_buyer_only": false,
  "income_limit_pct_ami": null,
  "special_conditions": [
    "Gift funds permitted up to 100% of down payment for primary residence",
    "Reserves required: 2 months PITI for SFR, 6 months for investment"
  ],
  "rate_today": 6.875,
  "rate_last_updated": "2026-04-28T08:00:00Z",
  "estimated_closing_costs_pct": 2.5,
  "link_lo_assigned": ["lo_id_1", "lo_id_2"]
}
```

Matching engine logic (pseudocode):

MATCHING ENGINE — PSEUDOCODE

```

function matchPrograms(consumer, programs):
  matches = []
  for program in programs:
    if consumer.fico < program.min_fico: continue
    if consumer.dti > program.max_dti: continue
    if consumer.ltv > program.max_ltv: continue
    if consumer.state not in program.states_eligible: continue
    if consumer.loan_amount < program.loan_amount_min: continue
    if consumer.loan_amount > program.loan_amount_max: continue
    if program.first_time_buyer_only and not consumer.is_fthb: continue
    matches.append({
      program: program,
      estimated_rate: program.rate_today + risk_adjustment(consumer),
      estimated_payment: pmt(consumer.loan_amount, rate, program.term_years),
      pmi_applies: consumer.ltv > program.pmi_required_below_ltv,
      conditions: program.special_conditions
    })
  return rank(matches, by=[lowest_payment, then_consumer_preference])

```

15.2 preQualRecord document — proposed shape**PREQUALRECORD — SAMPLE**

```

{
  "_id": "prequal_2026_04_28_abc123",
  "agent_id": "agent_xyz",
  "source_link_id": "link_listing_25586229",
  "consumer": {
    "name": "...", "email": "...", "phone": "...",
    "dob_year": 1985, "current_address": "..."
  },
  "loan_purpose": "purchase",
  "subject_property": { "address": "...", "estimated_value": 1400000 },
  "income": { "monthly_gross": 14500, "employment_type": "W2", "employer": "..." },
  "assets": { "cash_for_dp": 70000, "sources": ["savings", "gift"] },

```

```

"debts": { "monthly_total": 850 },
"credit": {
  "consent_timestamp": "...", "consent_ip": "...",
  "soft_pull_provider": "isoftpull",
  "fico": 742, "tradeline_summary": "...",
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15.3 Glossary

Term	Definition
1003	The standard residential mortgage loan application form (Fannie Mae form 1003 / Freddie Mac form 65). The dataset captured during pre-qualification.
A2A	Agent-to-agent. The protocol governing communication between specialized AI agents in our multi-agent reasoning architecture.
Arive	Loan origination system used by Link's mortgage operation. Integrates via Zapier only.
DTI	Debt-to-income ratio. Total monthly debt obligations divided by monthly gross income.
FCRA	Fair Credit Reporting Act. Federal law governing credit pull consent and consumer disclosures.
GLBA	Gramm-Leach-Bliley Act. Federal law governing the protection of consumer financial information.
LO	Loan officer. Link employee who closes mortgages.
LTV	Loan-to-value ratio. Loan amount divided by property value.
MLS	Multiple Listing Service. The agent-controlled database of active and sold

Term	Definition
	listings.
Net sheet	A seller's estimated proceeds at a given sale price after all fees and payoffs.
POS	Point of sale. The borrower-facing portion of a loan origination system.
Pre-qualification	A non-binding lender estimate of how much a buyer could borrow, based on consumer-reported data plus a soft credit pull.
RESPA	Real Estate Settlement Procedures Act. Federal law governing settlement-services referrals.
Soft pull	A credit inquiry that does not affect the consumer's credit score and does not require SSN.
Tri-merge	A combined credit report drawing from all three bureaus (Experian, Equifax, TransUnion).
White-label	Branding the consumer-facing surface as the agent's own rather than as Link.

DOCUMENT END

End of specification — version 1.0. Next deliverable is the wireframes and Figma alignment for Phase 1 features (pre-qual flow, net sheet flow, agent dashboard updates), to be produced after engineering review of this document.